SKILLPROOF

Don't just claim your skills. Prove them.

AI-powered evidence-based skill verification & talent matching

CANDIDATE → EVIDENCE → AI ANALYSIS → SKILL PROFILE → COMPANY MATCH

From Resume Claims → Real Evidence → Better Hiring

Project at a glance

SkillProof helps candidates demonstrate their abilities with supporting evidence and helps companies find suitable people based on skills, evidence strength, experience domain, and role requirements.

Core idea

A skill is not treated as a keyword alone. SkillProof links it to evidence such as projects, GitHub activity, documentation, certificates, experience, and practical assessments.

Core loop

Person → Evidence → AI Skill Analysis → Evidence Tier → Company Requirements → Explainable Matching → Hiring

This document is a concise project brief for team alignment and hackathon preparation—not a slide deck.

1. Problem, Real Gap & Solution

The problem

Resumes describe what candidates claim to know, but do not reliably demonstrate what they can do. Recruiters spend time filtering applications and verifying whether candidates have the skills required for a role.

The real gap

Current hiring	What is missing?
Resume → Keywords → Shortlist → Interview	Evidence that supports each skill claim
Claim ≠ demonstrated ability	A clear, explainable link between skill and proof

Our solution: SkillProof

An AI-powered evidence layer that analyzes candidate skill claims and their supporting material, creates an explainable skill profile, strengthens it with practical verification, and matches candidates to company requirements.

Positioning

We are not trying to replace job portals. We add an evidence layer between a candidate's claims and a company's confidence.

2. How SkillProof Works

Candidate journey

1. Submit evidence	Resume, GitHub, projects, documentation, certificates, and experience.
2. Extract skills	AI identifies claimed skills and connects them to relevant evidence sources.
3. Build the evidence graph	Each skill is linked to projects, code activity, documentation, and assessments.
4. Verify in practice	A role-specific challenge can test whether the candidate can apply the skill.
5. Create the evidence profile	Show evidence strength, sources, and gaps—without claiming absolute expertise.
6. Match to a role	Compare the profile with company requirements and explain the match.

Example journey

Aarav claims Python and Machine Learning. SkillProof reviews his project and GitHub evidence, offers a Python challenge, then shows which skills have strong or moderate supporting evidence. A company can see the evidence before considering him for a Junior AI Developer role.

Keep the system honest: evidence tiers are assessments of available evidence, not a guarantee of a person’s total ability.

3. Killer Features & Practical Skill Verification

Killer features

Claim vs Evidence	Compare what the candidate says with the evidence available for that skill.
Skill Evidence Graph	Connect each skill to projects, GitHub, documentation, certificates, and assessments.
Practical Skill Challenges	Offer short, role-specific tasks to collect direct evidence of applied ability.
Explainable Evidence Tiers	Show Strong, Moderate, or Limited Evidence with supporting sources and gaps.
Experience-domain matching	Match by field, skills, evidence tier, and relevant experience.
Explainable company matching	Show why a candidate matches a role instead of displaying a score alone.

Practical verification example

Claim: Python — Advanced
Challenge: data processing, API/data handling, or problem-solving task
Evaluation: correctness, approach, code quality, efficiency, and explanation
Output: practical evidence added to the candidate's profile.

AI supports evaluation; it should not be presented as infallible. Assessment results and evidence sources should remain visible to candidates and recruiters.

4. Skill Evidence Graph & Company Requirements

Skill Evidence Graph

Instead of storing “Python — Advanced” as a standalone keyword, SkillProof connects Python to the sources that support the claim.

SKILL	LINKED EVIDENCE	PROFILE OUTPUT
Python	Project A • Project B • GitHub • Coding assessment	Strong evidence
C++	Project C • GitHub	Moderate evidence
Machine Learning	ML project • Model implementation • Documentation	Strong project evidence

Company requirements

A company defines the role, experience domain, required skills, minimum evidence tier, and experience range. SkillProof compares candidates against those criteria.

Junior Python Developer	**Required evidence**
Python	Strong
SQL	Moderate or higher
Git	Moderate or higher
API development	Strong
Experience	0–2 years

5. Explainable Candidate Matching

A match score should come with reasons. SkillProof should show how each requirement compares with the candidate’s evidence.

Role requirement	Aarav’s evidence	Result
Python	Projects + coding challenge	Strong
Machine Learning	ML projects + documentation	Strong project evidence
Git	Repository activity	Strong
SQL	One project; limited assessment evidence	Moderate
API development	Project evidence	Moderate

91% example evidence match

Why recommended? Strong evidence for Python, Machine Learning, and Git; moderate evidence for SQL and APIs.

Important

The percentage is an illustrative prototype example, not a validated hiring prediction. The system should explain its criteria, show evidence gaps, and leave the final decision to the company.

6. What Makes SkillProof Different?

Traditional approach	SkillProof approach
Resume claims	Claims connected to evidence
Keyword-based filtering	Evidence-based skill matching
Certificates as signals	Projects + code + assessments + experience
Generic screening	Role-specific practical challenges
Score with little context	Explainable match with requirement-by-requirement reasons

Potential impact

For candidates: clearer demonstration of skills, visibility into evidence gaps, and more ways to show practical ability.

For companies: more focused shortlisting, faster evidence review, and better visibility into role-specific capability.

Future scope

More assessment types • verified certificate integrations • organization-specific challenges • blind screening options • continuously updated skill profiles.

7. Technology, MVP & Tagline

Suggested technology stack

Layer	Possible technology	Purpose
Frontend	React / Vite	Candidate and company dashboards
Backend	FastAPI or Node.js / Express	APIs, workflow, matching
Database	Firebase or Supabase	Profiles, evidence, roles
AI	Gemini API	Skill extraction and explanations
Integrations	GitHub API + PDF/text extraction	Collect and parse evidence
Scoring	Rule-based / Python scoring engine	Transparent evidence tiers and match reasons

Hackathon MVP — build these first

1. Candidate profile and evidence upload
2. Skill extraction from sample resume/project data
3. Claim vs Evidence view with evidence tiers
4. One practical challenge for a selected skill
5. Company role form and explainable candidate matching

Safety and fairness principles

Use consented evidence, protect candidate data, show why a tier or match was assigned, avoid treating AI scores as absolute truth, and keep human hiring decisions in control.

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